



SWISS KNIFE

- Plug-and-Play Alignment Heads via Tournament Sampling Auditors in Speculative Decoding

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Abstract

Alignment today is still checkpoint-bound: most pipelines implicitly hard-code a single behavioral compromise into a trained model, making it costly to *revise*, *specialize*, or *hot-fix* behavior after deployment (e.g., stricter *safety*, higher *helpfulness*, stronger *harmlessness*, improved *informativeness*, or different refusal posture). While instruction-conditioned switching can increase controllability, it still assumes a **single frozen policy** must internalize *multiple behavioral regimes* and remain stable under interference and distribution shift. We take a step beyond frozen-backbone switching by **externalizing alignment into a changeable decode-time head**. We introduce **Swiss Knife**, a **protocol** that repurposes **speculative decoding** into an *alignment socket*: a fast *draft* model generates candidates, while a separately trainable *auditor* model enforces a *chosen alignment objective during generation*.

Swiss Knife’s key algorithmic component is the Tournament Sampling Auditor (TSA). At each decoding step (token-level) or short horizon (span-level), the draft model proposes a **candidate set** and the auditor conducts an *efficient tournament*—either a **pairwise knockout bracket** or a **Swiss-system** schedule—to select the winner under an objective-specific score. This yields **stable preference-based selection** while mitigating degenerate solutions such as *refuse-always* or templated “safe” boilerplate. Crucially, Swiss Knife supports **objective-specific auditor “blades”**—*safety*, *helpfulness*, *harmlessness*, *informativeness*, and style constraints—that can be **tuned, swapped, and updated independently** of the backbone, enabling rapid post-deployment alignment updates without retraining the generator. We formalize Swiss Knife as **decode-time policy shaping** that combines draft likelihood with auditor scores, and we propose an evaluation protocol that holds prompts fixed while varying objectives to quantify **switchability** (objective adherence and cross-objective separation) alongside **robustness** (harmful compliance, refusal correctness, uncertainty calibration) and **systems realism** (acceptance rate, auditor calls per token, latency and throughput). Across objectives and model scales, Swiss Knife delivers stronger multi-regime alignment under comparable compute budgets, demonstrating that **modular auditor heads** provide a *practical, updatable, and deployment-friendly* path to switchable alignment.